

Problem_Set_2

Group 2

2026-02-26

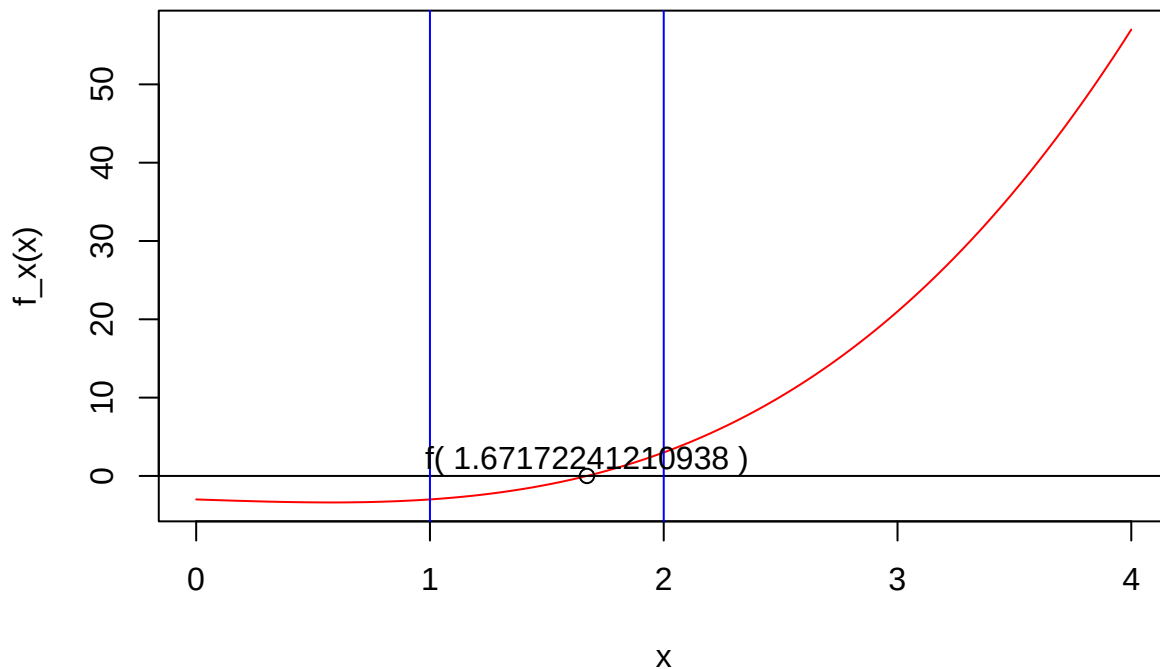
1.) Use the bisection method with a hand calculator or computer to find the real root of $x^3 - x - 3 = 0$. Use an error tolerance of $\epsilon = 0.0001$. Graph the function $f(x) = x^3 - x - 3$ and label the root.

Solution:

Finding the root using bisection, given $[a, b] = [1, 2]$.

[1] 1.671722

The root is 1.671722



2.) The function $f(x) = -3x^3 + 2e^{\frac{x^2}{2}} - 1$ has values of zero near $x = -0.5$ and $x = 0.5$.

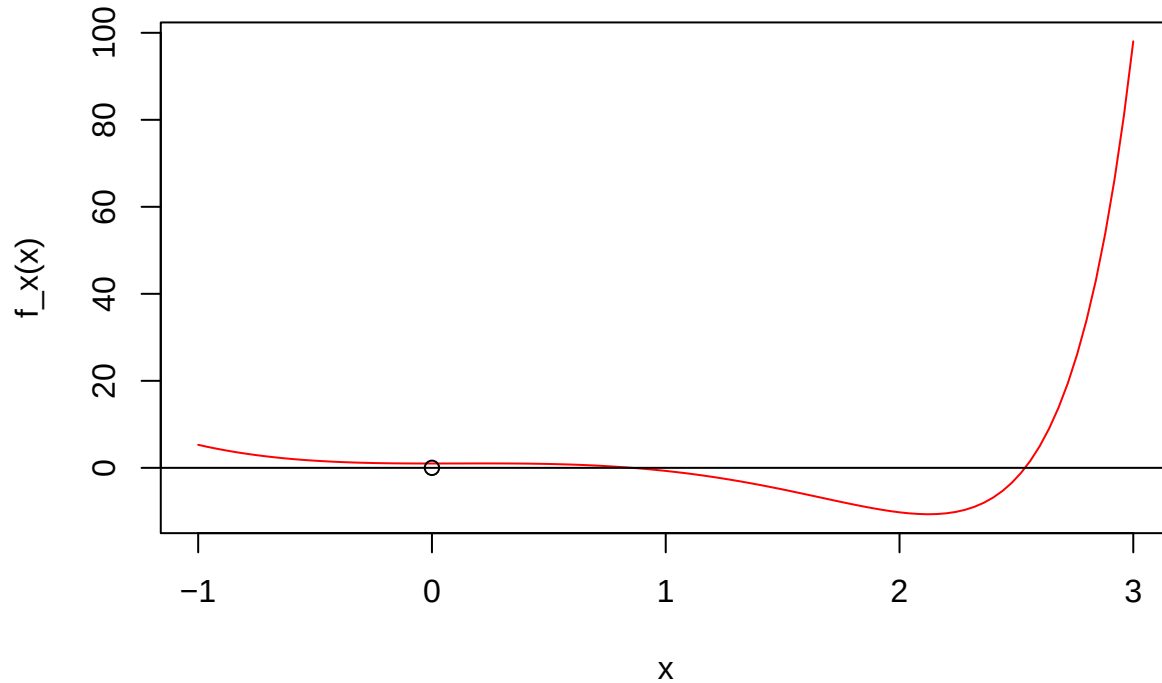
a. What is the derivative of f ?

Answer: $x(2e^{\frac{x^2}{2}} - 9x)$

b. If you begin Newton's method at $x = 0$, which root is reached? How many iterations to achieve an error less than 10^{-5} ?

Answer: If the x is equals to 0, a root is not reached since in solving for the next x using the equation: $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$ requires a denominator with nonzero value i.e. $f'(x) \neq 0$. As for the number of iterations, it's not applicable since it simply stops at the first iteration. The plot below plots $f'(0)$.

Plotting $f(x)$



c. Begin Newton's method at another starting point to get the other zero.

Answer: Using the starting point $x = 1$

```
## [1] 1
## [1] 0.8767996
## [1] 0.8572855
## [1] 0.8568009
## [1] 0.8568006
```

Therefore, the root is 0.8568006 when $x = 1$

3.) Use the function from no. 2 and find the root using the secant method where $x_0 = 0$ and $x_1 = 1$. Use an error tolerance of $\epsilon = 0.001$.

```
## [1] 0.5873517
## [1] 0.8029441
## [1] 0.882793
## [1] 0.8549891
## [1] 0.8567427
## [1] 0.8568007
```

Therefore, the root is 0.8568007 when $x_0 = 0, x_1 = 1$, and $\epsilon = 0.001$ using the Secant Method

4.) Consider the system of equations:

$$\begin{cases} 2x + 3y - z = 5 \\ 4x - y + 2z = 6 \\ -3x + 2y + z = -4 \end{cases}$$

a. Present the augmented matrix of the system. Answer:

$$\left(\begin{array}{ccc|c} 2 & 3 & -1 & 5 \\ 4 & -1 & 2 & 6 \\ -3 & 2 & 1 & -4 \end{array} \right)$$

b. Solve the system using $Ax = LUx = Ly = b$ and round the final answer to 4 decimal digits.

```
## [1] "Row Operation: R_2 ← R_2 - (2.00)R_1"
## [1] "Upper:"
##      [,1] [,2] [,3]
## [1,]    2    3   -1
## [2,]    0   -7    4
## [3,]   -3    2    1
## [1] "Lower:"
##      [,1] [,2] [,3]
## [1,]    1    0    0
## [2,]    2    1    0
## [3,]    0    0    1
## [1] "Row Operation: R_3 ← R_3 - (-1.50)R_1"
## [1] "Upper:"
##      [,1] [,2] [,3]
## [1,]    2  3.0 -1.0
## [2,]    0 -7.0  4.0
## [3,]    0  6.5 -0.5
## [1] "Lower:"
##      [,1] [,2] [,3]
## [1,]   1.0    0    0
## [2,]   2.0    1    0
## [3,]  -1.5    0    1
## [1] "Row Operation: R_3 ← R_3 - (-0.93)R_2"
## [1] "Upper:"
##      [,1] [,2]      [,3]
## [1,]    2    3 -1.000000
## [2,]    0   -7  4.000000
## [3,]    0    0  3.214286
## [1] "Lower:"
##      [,1]      [,2] [,3]
## [1,]   1.0  0.0000000  0
## [2,]   2.0  1.0000000  0
## [3,]  -1.5 -0.9285714  1

##      [,1]
## [1,]  5.0000
## [2,] -4.0000
## [3,] -0.2143

##      [,1]
## [1,]  1.6667
## [2,]  0.5333
## [3,] -0.0667
```

Therefore, the final answer is $x = 1.6667, y = 0.5333, z = -0.0667$

c. Find the residual vector if the correct solution is: $x = 1, y = 2, z = -1$ Answer:

The residual vector can be solved by finding the difference between the correct solution and the answers from the previous question:

$$\begin{aligned}x_{residual} &= 1 - 1.6667 = -0.6667 \\y_{residual} &= 2 - 0.5333 = 1.4667 \\z_{residual} &= -1 - (-0.0667) = -0.9333\end{aligned}$$

5.) Compute the Frobenius norm, maximum column sum, and maximum row sum of the matrix:

$$\begin{pmatrix} 10.2 & 2.4 & 4.5 \\ -2.3 & 7.7 & 11.1 \\ -5.5 & -3.2 & 0.9 \end{pmatrix}$$

Answer:

6.) Solve the system of equations given in no.4, starting with the initial vector of $[0, 0, 0]$:

- a. Solve using the Jacobi method with 2-digit precision.
- b. Solve using Gauss-Seidel method with 2-digit precision.
- c. Solve for $\bar{e}$ if the true solution is: $x = 1, y = 2, z = -1$